

Grammarcetamol — Student Frontend Architecture & Design Specification

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Application: Grammarcetamol-Student-Interface
Stack: Next.js (App Router), TypeScript, Tailwind CSS, React Query, Zustand
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1. Overview

The **Student Interface** is the public-facing, learner-centric application of the Grammarcetamol ecosystem. It is designed to be the primary touchpoint for students discovering, purchasing, and consuming English language learning content. Every design decision prioritizes **Learning First** — reducing cognitive load, maximizing engagement, and ensuring seamless access to educational material across all devices.

1.1 Design Philosophy

- **Clarity over decoration:** Every element serves a learning purpose.
- **Progressive disclosure:** Complex features reveal themselves only when needed.
- **Responsive-first:** Native-app quality experience on mobile, tablet, and desktop.
- **Accessibility-first:** WCAG 2.1 AA compliance is non-negotiable.
- **Performance-critical:** Pages must feel instant; videos must start within 2 seconds.

1.2 Target Users

User Type	Access Level	Primary Goal
Guest	Unauthenticated	Discover, evaluate, and register
Registered Student	Authenticated	Learn, track progress, engage

2. Information Architecture

```
grammarcetamol-student-interface/  
├─ Public Routes (Guest + Student)  
│   ├── /                               → Landing Page (Hero, Services, Courses, Testimonials)  
│   ├── /courses                       → Course Catalog (Filter, Search, Pagination)  
│   ├── /courses/[slug]                → Course Detail (Overview, Curriculum, Reviews, Instructor)  
│   ├── /services                      → Services Directory (7 Flagship Services)  
│   ├── /services/[slug]               → Service Detail (Benefits, Pricing, Request Form)  
│   ├── /instructors                   → Instructor Directory  
│   ├── /instructors/[id]              → Instructor Profile  
│   ├── /blog                          → Blog/Articles Listing  
│   ├── /blog/[slug]                   → Article Detail  
│   ├── /about                         → About Grammarcetamol  
│   └── /faq                           → Frequently Asked Questions
```

		/contact	→ Contact & Enquiry Form
		/testimonials	→ Student & Corporate Testimonials
		/login	→ Authentication (Email / Google OAuth)
		/register	→ Registration (Email / Google OAuth)
		/forgot-password	→ Password Reset Flow
		/verify-email	→ Email Verification Handler
		Protected Routes (Student Only)	
		/dashboard	→ Student Learning HQ
		/my-courses	→ Enrolled Courses (Purchased, Free, Completed)
		/my-courses/[courseId]	→ Course Learning Interface
		/live-classes	→ Upcoming & Past Live Classes
		/live-classes/[id]/join	→ Live Class Join Portal
		/coaching	→ One-on-One Coaching Booking
		/coaching/[bookingId]	→ Booking Confirmation & Details
		/services/request	→ Service Request Form (Authenticated)
		/notifications	→ Notification Center
		/profile	→ Profile Settings
		/profile/edit	→ Edit Profile & Preferences
		/transactions	→ Payment & Transaction History
		/certificates	→ Earned Certificates (Future)
		/settings	→ Account & Privacy Settings
		Checkout Flow	
		/checkout/[courseId]	→ Branded Checkout Page
		/checkout/success	→ Payment Confirmation

3. Global Design System

3.1 Color Palette

Token	Hex	Usage
--color-primary	#1E3A5F	Primary brand color, headers, key actions
--color-primary-light	#2A5285	Hover states, secondary emphasis
--color-accent	#F59E0B	CTAs, progress indicators, highlights
--color-accent-hover	#D97706	CTA hover states
--color-success	#10B981	Success states, completion, confirmations
--color-warning	#F59E0B	Warnings, pending states
--color-error	#EF4444	Errors, failures, destructive actions
--color-background	#F8F8F8	Page backgrounds
--color-surface	FFFFFF	Cards, modals, elevated surfaces
--color-surface-elevated	FFFFFF + shadow	Dropdowns, popovers
--color-border	#E2E8F0	Dividers, input borders

Table 2 – continued

Token	Hex	Usage
--color-text-primary	#0F172A	Headings, primary text
--color-text-secondary	#64748B	Body text, descriptions
--color-text-muted	#94A3B8	Placeholders, disabled text

3.2 Typography

Element	Font	Weight	Size (Desktop)	Size (Mobile)	Line Height
H1 (Hero)	Inter	800	56px	36px	1.1
H2 (Section)	Inter	700	40px	28px	1.2
H3 (Card Title)	Inter	600	24px	20px	1.3
H4 (Subsection)	Inter	600	20px	18px	1.4
Body	Inter	400	16px	15px	1.6
Body Small	Inter	400	14px	13px	1.5
Caption	Inter	500	12px	11px	1.4
Button	Inter	600	16px	15px	1.0
Nav Link	Inter	500	15px	14px	1.0

3.3 Spacing Scale

Based on an 8px grid system:

Token	Value
space-1	4px
space-2	8px
space-3	12px
space-4	16px
space-5	24px
space-6	32px
space-7	48px
space-8	64px
space-9	96px
space-10	128px

3.4 Border Radius

Token	Value	Usage
radius-sm	4px	Buttons, badges, small elements
radius-md	8px	Inputs, cards, modals
radius-lg	12px	Large cards, feature sections
radius-xl	16px	Hero containers, media players
radius-full	9999px	Pills, avatars, circular buttons

3.5 Shadows

Token	Value	Usage
shadow-sm	0 1px 2px rgba(0,0,0,0.05)	Subtle elevation
shadow-md	0 4px 6px -1px rgba(0,0,0,0.1)	Cards, dropdowns
shadow-lg	0 10px 15px -3px rgba(0,0,0,0.1)	Modals, popovers
shadow-xl	0 20px 25px -5px rgba(0,0,0,0.1)	Overlays, toasts

3.6 Animation Tokens

Token	Duration	Easing	Usage
transition-fast	150ms	ease-out	Button hovers, toggles
transition-base	200ms	ease-in-out	Card hovers, dropdowns
transition-slow	300ms	ease-in-out	Page transitions, modals
transition-slower	500ms	cubic-bezier(0.4, 0, 0.2, 1)	Hero animations

4. Component Inventory

4.1 Layout Components

Navbar

- **Position:** Fixed top, z-index 50, height 72px
- **Behavior:** Transparent on hero → solid white on scroll (trigger: 50px)
- **Elements:**
 - Logo (left)
 - Navigation links: Courses, Services, Live Classes, Blog, About (center, hidden on mobile)
 - Search icon → expands to search bar
 - Notification bell (student only, with unread badge)
 - User avatar dropdown (student) or Login/Register buttons (guest)
- **Mobile:** Hamburger menu → full-screen overlay navigation

Footer

- **Layout:** 4-column grid (desktop), stacked (mobile)
- **Columns:**
 1. Brand + short description + social icons
 2. Quick Links (Courses, Services, About, Blog, FAQ)
 3. Legal (Privacy Policy, Terms, Refund Policy)
 4. Newsletter signup (email input + subscribe button)
- **Bottom bar:** Copyright + payment method icons

PageHeader

- **Usage:** Section headers across internal pages
- **Structure:** Breadcrumb trail → Title → Optional subtitle/description → Optional action button
- **Styling:** Background `--color-background`, bottom border `--color-border`, padding `space-7` vertical

Sidebar (Learning Interface)

- **Position:** Fixed left, width 320px (desktop), collapsible drawer (mobile)
- **Structure:**
 - Course title + progress ring
 - Module accordion list
 - Each module: title + completion percentage + lesson list
 - Lesson item: status icon (locked/playing/completed) + title + duration
 - Active lesson highlighted with `--color-primary` left border

4.2 Data Display Components

CourseCard

- **Dimensions:** 280px min-width, aspect-ratio 16:9 thumbnail
- **Elements:**
 - Thumbnail image with hover zoom (scale 1.05)
 - Difficulty badge (top-left)
 - Price badge (top-right, or “Free” label)
 - Title (2-line clamp)
 - Instructor name + avatar
 - Rating stars + count
 - Student count
- **Hover:** Shadow elevation increase, “Preview” overlay on thumbnail
- **States:** Loading skeleton, empty state, error state

StatCard (Dashboard)

- **Structure:** Icon (48px, rounded container) + Value (H3) + Label (caption) + Trend indicator (up/down arrow + percentage)
- **Variants:** Primary (colored background), Default (white background)

ProgressRing

- **Usage:** Course completion, module progress
- **Specs:** 48px diameter, 4px stroke, `--color-accent` for progress, `--color-border` for track
- **Center:** Percentage text (caption size)

NotificationItem

- **Structure:** Icon (colored by type) + Title + Message preview + Timestamp + Unread dot
- **Types:** Course (blue), Payment (green), Live Class (purple), Announcement (orange), System (gray)
- **Actions:** Mark as read, Delete, Click to navigate

4.3 Form Components

Input

- **States:** Default, Focus (ring-2 --color-primary), Error (ring-2 --color-error), Disabled
- **Features:** Label, helper text, error message, icon prefix/suffix, password visibility toggle

SearchInput

- **Behavior:** Debounced (300ms), loading spinner during fetch, dropdown results with course thumbnails
- **Keyboard:** Arrow navigation, Enter to select, Escape to close

ResourceDownloader

- **Usage:** Download supplementary course materials (PDFs, worksheets, audio files)
- **Visual:** File list with icon (by type), file name, size, and download button
- **States:** Available (clickable download), Downloading (progress indicator), Downloaded (checkmark + “Open” link), Restricted (lock icon for premium resources)
- **Security:** Signed URL generation for each download, expiring after 15 minutes

4.4 Feedback Components

Toast

- **Position:** Bottom-right (desktop), top-center (mobile)
- **Types:** Success, Error, Warning, Info
- **Duration:** 4 seconds (auto-dismiss), persistent for errors
- **Animation:** Slide in from right, fade out

Modal

- **Overlay:** Backdrop blur + dark overlay (rgba(0,0,0,0.5))
- **Sizes:** Small (400px), Medium (560px), Large (720px), Fullscreen (mobile)
- **Close:** X button, Escape key, click outside
- **Focus trap:** First focusable element auto-focused

Skeleton

- **Usage:** Loading states for cards, lists, text blocks
- **Style:** Animated shimmer gradient (--color-border to --color-background)

4.5 Media Components

VideoPlayer

- **Library:** Custom wrapper around HTML5 video + hls.js for adaptive streaming
- **Controls:** Play/Pause, Seek bar with buffered indicator, Volume, Playback speed (0.5x–2x), Fullscreen, Picture-in-Picture, Captions toggle
- **Overlay:** Title, next lesson preview (last 10 seconds)
- **Keyboard:** Space (play/pause), ArrowLeft/Right (seek ± 10 s), ArrowUp/Down (volume), F (fullscreen), M (mute)
- **Progress sync:** Auto-save current time to backend every 5 seconds

5. Page Specifications

5.1 Landing Page (/)

Objective: Convert visitors into registered students within 60 seconds.

Sections (top to bottom):

1. Hero Section

- Full viewport height (100vh), centered content
- Background: Subtle animated gradient or high-quality educational imagery (overlay: `--color-primary` at 85% opacity)
- Headline: “Master English with Confidence” (H1, white)
- Subheadline: 2-line description of platform value (Body Large, white/80%)
- CTAs: Primary “Start Learning” (accent button, large) + Secondary “Explore Courses” (outline button, white)
- Trust indicators: “10,000+ students”, “4.9/5 rating”, “Expert instructors” (horizontal row below CTAs)
- Scroll indicator: Animated chevron at bottom

2. Services Section

- Section title: “Our Flagship Services” (H2)
- Grid: 3 columns desktop, 1 column mobile
- 7 service cards: Professional illustration (120px) + Title (H4) + 2-line description + “Learn More” link
- Cards: White background, border, hover lift + shadow

3. Featured Courses Section

- Section title: “Popular Courses” (H2) + “View All →” link
- Horizontal scroll carousel (desktop: 4 visible, mobile: 1.5 visible with peek)
- CourseCard components
- Auto-scroll disabled; manual scroll with momentum

4. How It Works

- 3-step horizontal timeline: Register → Learn → Master
- Each step: Numbered circle + Icon + Title + Description
- Connecting line between steps (desktop)

5. Testimonials

- Section title: “What Our Students Say”
- Carousel: Large quote text + Student photo (64px circle) + Name + Occupation
- Auto-advance every 6 seconds, manual dots navigation
- 5 testimonials minimum for credibility

6. FAQ Section

- Accordion layout: Question (bold) → Expandable answer
- 8–10 questions covering courses, payments, live classes, certificates, refunds, tech requirements
- Smooth height animation on expand/collapse

7. CTA Banner

- Full-width, `--color-primary` background
- Text: “Ready to Transform Your English?” (H2, white)
- Button: “Get Started for Free” (accent, large)

5.2 Course Catalog (/courses)

Layout: Sidebar filters (left, 280px) + Course grid (right)

Sidebar Filters:

- Search bar (sticky top)
- Categories (checkbox list with count badges)
- Difficulty (radio: Beginner, Intermediate, Advanced, All)
- Price (radio: Free, Paid, All)
- Rating (checkbox: 4 & up, 3 & up)
- Duration (range slider)
- Clear Filters button

Main Area:

- Result count + Sort dropdown (Popular, Newest, Price Low-High, Price High-Low, Highest Rated)
- Course grid: 3 columns desktop, 2 tablet, 1 mobile
- Pagination: Load more button (infinite scroll optional)
- Empty state: Illustration + “No courses match your filters” + Clear filters CTA

5.3 Course Detail (/courses/[slug])

Layout: Two-column (left 65%, right 35% desktop), stacked mobile

Left Column:

- Breadcrumb: Home > Courses > [Category] > [Title]
- Video thumbnail with play overlay (promo video) or course trailer
- Title (H1) + Subtitle (Body Large, secondary)
- Meta row: Rating stars + count | Student count | Last updated | Language
- Tabs: Overview | Curriculum | Reviews | Instructor
- **Overview:** Description, learning objectives (checklist), prerequisites, target audience
- **Curriculum:** Module accordion → Lesson list with duration and preview icons
- **Reviews:** Aggregate rating breakdown (1-5 star bars) + Review cards (pagination)
- **Instructor:** Bio, photo, stats (courses, students, rating)

Right Column (Sticky):

- Price card: Original price (strikethrough) + Discounted price + “Buy Now” button (full width, accent)
- “Enroll for Free” button (if free)
- Includes: Lifetime access, Certificate of completion, Mobile access
- 30-Day Money-Back Guarantee badge

- Share buttons

Bottom: Related courses carousel

5.4 Student Dashboard (/dashboard)

Layout: Full-width content area, no sidebar (top nav only)

Sections:

1. Welcome Card

- Full-width banner, --color-primary gradient background
- Left: “Good morning, [First Name]! 🙌” (H3, white) + Motivational subtext + Date
- Right: Learning streak flame icon + “5-day streak” (future) or “Start your streak today!”

2. Continue Learning

- Horizontal card: Course thumbnail (left, 200px) + Course title + Progress bar + “Resume” button + “X% Complete”
- If no active course: “You haven’t started a course yet” + Browse courses CTA

3. My Courses Grid

- Tabs: In Progress | Completed | Not Started
- 3-column grid of mini course cards (thumbnail + title + progress bar + last accessed)
- “View All” link to /my-courses

4. Upcoming Live Classes

- Horizontal scroll list
- Card: Date badge (day + month) + Title + Time + Instructor avatar + “Join” button (enabled 15 min before) or countdown
- Empty state: “No upcoming classes” + Browse live classes link

5. Notifications Panel

- Latest 5 notifications with “View All” link
- Inline mark-as-read functionality

6. Recommended Courses

- “Because you enrolled in [Course]” or “Trending Now”
- Horizontal scroll of CourseCard components

5.5 Learning Interface (/my-courses/[courseId])

Layout: Three-pane layout (desktop), stacked (mobile)

Left Sidebar (320px, collapsible):

- Course header: Thumbnail (40px) + Title (truncated) + Overall progress ring
- Module list (accordion):
- Module title + expand/collapse chevron + completion fraction (e.g., “2/5”)
- Lesson items:
- Icon: Lock (locked) | Play circle (available) | Check circle (completed) | Pulse (currently playing)
- Title + Duration
- Active lesson: --color-primary left border, subtle background highlight
- Resources section at module bottom (if any)

Main Content Area (flexible):

- Video player (16:9, max-height 70vh)
- Below video:
- Lesson title (H3) + Completion checkbox (“Mark as Complete”)
- Lesson notes/description (rich text)
- Resources list (downloadable files)
- Navigation: “← Previous Lesson” | “Next Lesson →”

Right Sidebar (280px, collapsible, desktop only):

- Instructor mini-card: Avatar + Name + “View Profile” link
- Downloads: File list with size + download icon
- Discussion: Recent comments preview + “Open Discussion” button
- Bookmarks: Toggle bookmark for current timestamp
- My Notes: Quick-add text area (auto-saved)

Mobile Adaptation:

- Left sidebar becomes bottom sheet (swipe up)
- Right sidebar becomes tabs below video: Notes | Discussion | Downloads
- Video player supports fullscreen rotation

5.6 Branded Checkout (/checkout/[courseId])

Layout: Two-column (left: order summary, right: payment form)

Left Column (Order Summary):

- Course thumbnail (120px) + Title + Instructor
- Price breakdown: Subtotal | Discount (if any) | Tax (if applicable) | **Total**
- Secure payment badges (SSL, PCI DSS)
- Money-back guarantee reminder

Right Column (Payment Form):

- Customer details (pre-filled from profile, editable): Full name, Email, Phone
- Payment method selector: Card (default), Mobile Money, Bank Transfer (configurable per region)
- Card form: Number, Expiry, CVV, Name on card (if card selected)
- “Pay [Amount]” button (full width, accent, loading state during processing)
- Terms agreement checkbox

Success State:

- Full-screen confirmation: Checkmark animation + “Payment Successful!”
- Course access card: “Start Learning Now” button + “Go to Dashboard” link

- Email confirmation notice

Failure State:

- Inline error message with specific reason (e.g., “Insufficient funds”, “Card declined”)
- “Try Again” button + “Change Payment Method” option

5.7 Live Class Portal (/live-classes & /live-classes/[id]/join)

Listing Page (/live-classes):

- Tabs: Upcoming | Past
- Filter: Date range, Category, Instructor
- Card: Banner image + Title + Date/Time (with timezone) + Duration + Instructor + Price/Free badge + Capacity indicator (e.g., “12/30 spots left”) + Register/Join button

Join Page (/live-classes/[id]/join):

- Pre-join screen (5 min before start): Countdown timer + Class details + “Waiting for host...” + Tech check (camera/mic test if integrated)
- Active state: Embedded video conference (Jitsi/Zoom SDK/custom WebRTC) or external link launcher
- Post-class: Recording availability notice (if applicable) + “Rate this class” prompt

5.8 Profile & Settings (/profile)

Tabs: Profile | Account | Notifications | Privacy

Profile Tab:

- Avatar upload (drag-drop or click, max 2MB, preview crop)
- Form fields: Full name, Email (read-only if verified), Phone, Country (dropdown), Timezone (auto-detect + manual), Biography (textarea, 500 chars), Learning goals (tag input)
- “Save Changes” button (disabled until dirty)

Account Tab:

- Change password: Current | New | Confirm
- Linked accounts: Google (connect/disconnect)
- Danger zone: Delete account (requires confirmation + password)

Notifications Tab:

- Toggle list: Course updates, Live class reminders, Payment confirmations, Marketing emails, New course releases
- Delivery preference per type: In-app only | Email | Both

Privacy Tab:

- Profile visibility: Public | Students only | Private
- Activity sharing: Show courses on profile | Show achievements

- Data export: “Download my data” (GDPR compliance)
-

6. State Management

6.1 Global State (Zustand)

```
// stores/authStore.ts
interface AuthState {
  user: Student | null;
  isAuthenticated: boolean;
  isLoading: boolean;
  login: (credentials: LoginDTO) => Promise<void>;
  logout: () => void;
  updateProfile: (data: ProfileUpdateDTO) => Promise<void>;
}

// stores/cartStore.ts (future)
interface CartState {
  items: CourseItem[];
  addItem: (course: Course) => void;
  removeItem: (courseId: string) => void;
  clearCart: () => void;
  total: number;
}

// stores/uiStore.ts
interface UIState {
  sidebarOpen: boolean;
  theme: 'light' | 'dark' | 'system';
  toasts: Toast[];
  addToast: (toast: Omit<Toast, 'id'>) => void;
  removeToast: (id: string) => void;
}
```

6.2 Server State (React Query / TanStack Query)

```
// hooks/useCourses.ts
export const useCourses = (filters: CourseFilters) =>
  useQuery({
    queryKey: ['courses', filters],
    queryFn: () => fetchCourses(filters),
    staleTime: 5 * 60 * 1000, // 5 minutes
  });

// hooks/useCourseProgress.ts
export const useCourseProgress = (courseId: string) =>
  useQuery({
    queryKey: ['progress', courseId],
    queryFn: () => fetchProgress(courseId),
    refetchInterval: 30000, // Poll every 30s for sync
  });
```

```

});

// hooks/useNotifications.ts
export const useNotifications = () =>
  useQuery({
    queryKey: ['notifications'],
    queryFn: fetchNotifications,
    refetchInterval: 60000,
  });

```

6.3 Local State

- Form inputs (React Hook Form)
- UI toggles (sidebar, modals, dropdowns)
- Video player state (currentTime, playbackRate, volume)
- Video playback state (currentTime, playbackRate, volume, buffering)

7. API Integration Patterns

7.1 Authentication Flow

1. User submits credentials → POST /api/auth/login
2. Backend returns JWT (access + refresh tokens)
3. Access token stored in httpOnly cookie (or memory)
4. Refresh token stored in httpOnly cookie
5. Axios interceptor attaches Bearer token to requests
6. On 401: Attempt silent refresh → retry request or redirect to login
7. On refresh failure: Clear auth state → redirect to /login with returnUrl

7.2 Video Streaming & Resource Delivery

1. Student opens lesson → GET /api/courses/{courseId}/lessons/{lessonId}
2. Backend returns lesson metadata including:
 - videoUrl (HLS manifest or direct MP4)
 - resources[] (signed download URLs)
 - lastWatchedPosition (timestamp in seconds)
3. Video Player loads HLS manifest → Adaptive bitrate streaming
4. Progress tracking:
 - Debounced PATCH /api/progress every 5 seconds
 - Payload: { lessonId, currentTime, completed: boolean }
5. Resource download:
 - Student clicks download → GET /api/resources/{resourceId}/download
 - Backend generates signed URL (15-min expiry)
 - Browser initiates download
6. Resume on return:
 - GET /api/progress/{courseId} → Returns last position per lesson
 - Video player seeks to timestamp automatically

7.3 Real-Time Features (Future)

- **Live Class:** WebRTC or embedded Zoom/Jitsi SDK
- **Notifications:** Server-Sent Events (SSE) or WebSocket for instant delivery
- **Progress Sync:** Debounced PATCH requests (every 5s) + optimistic UI updates

8. Responsive Breakpoints

Name	Width	Target
xs	< 480px	Small mobile phones
sm	480px -767px	Large mobile phones
md	768px -1023px	Tablets
lg	1024px -1279px	Small laptops
xl	1280px -1535px	Desktops
2xl	1536px	Large monitors

Key Responsive Behaviors

- **Navbar:** Full links (lg+) → Hamburger (md and below)
- **Course Grid:** 4 columns (xl) → 3 (lg) → 2 (md) → 1 (sm)
- **Learning Interface:** 3-pane (lg+) → Sidebar drawers (md) → Bottom sheets (sm)
- **Dashboard:** Multi-column grids collapse to single column below md
- **Font sizes:** Scale down 15-25% on mobile per typography scale

9. Accessibility Requirements

9.1 Keyboard Navigation

- All interactive elements reachable via Tab
- Logical tab order (top-to-bottom, left-to-right)
- Escape closes modals, dropdowns, and sidebars
- Space/Enter activates buttons and links
- Arrow keys navigate within menus, tabs, and carousels

9.2 Screen Reader Support

- Semantic HTML5 elements (<nav>, <main>, <article>, <aside>)
- ARIA labels for icon-only buttons
- aria-expanded for accordions and dropdowns
- aria-live="polite" for toast notifications and dynamic content
- Skip-to-content link for keyboard users

9.3 Visual Accessibility

- Minimum contrast ratio: 4.5:1 for normal text, 3:1 for large text
- Focus indicators: 2px solid outline, offset 2px, --color-primary
- No information conveyed by color alone (icons + text for status)

- Respect **prefers-reduced-motion**: Disable animations, instant transitions

9.4 Video Accessibility

- All videos include captions (VTT format)
- Transcripts available below video player
- Keyboard controls for all player functions
- High-contrast control icons

10. Performance Budgets

Metric	Target	Maximum
First Contentful Paint (FCP)	< 1.0s	1.5s
Largest Contentful Paint (LCP)	< 2.0s	2.5s
Time to Interactive (TTI)	< 2.5s	3.5s
Cumulative Layout Shift (CLS)	< 0.05	0.1
Total Blocking Time (TBT)	< 200ms	350ms
First Input Delay (FID)	< 50ms	100ms
Lighthouse Performance Score	95	90

Optimization Strategies

- **Images:** Next.js `<Image>` with WebP/AVIF, lazy loading, blur placeholder
- **Videos:** HLS adaptive streaming, poster images, lazy load below fold
- **Code:** Route-based code splitting, tree shaking, dynamic imports for heavy components
- **Fonts:** `font-display: swap`, preload critical font weights
- **Caching:** SWR for server state, localStorage for user preferences
- **CDN:** Static assets served via edge CDN

11. Error Handling & Empty States

11.1 Global Error Boundary

- Catches React rendering errors
- Displays friendly “Something went wrong” page
- “Reload page” button + “Contact support” link
- Error logging to monitoring service (Sentry)

11.2 Network Error Patterns

- **Retry:** Auto-retry failed GET requests (3 attempts, exponential backoff)
- **Offline:** Service Worker caches critical pages; “You are offline” banner
- **Timeout:** Requests timeout after 10s; show “Taking longer than expected...”

11.3 Empty States

Context	Visual	Message	CTA
No courses enrolled	Empty folder illustration	“You haven’ t enrolled in any courses yet.”	“Browse Courses”
No search results	Magnifying glass illustration	“No courses match your search.”	“Clear Filters”
No notifications	Bell illustration	“You’ re all caught up!”	—
No live classes	Calendar illustration	“No upcoming live classes.”	“View Schedule”
Cart empty	Shopping bag illustration	“Your cart is empty.”	“Explore Courses”

12. Third-Party Integrations

Service	Purpose	Integration Point
Stripe / Paystack / Flutterwave	Payment processing	Checkout page, webhook handlers
Google OAuth 2.0	Social authentication	Login/Register modal
Cloudinary / AWS S3	Media storage & delivery	Video streaming, image optimization
Jitsi Meet / Zoom SDK	Live video classrooms	Live class join page
SendGrid / AWS SES	Transactional emails	Backend-triggered (welcome, receipts)
Sentry	Error monitoring	Global error boundary
Google Analytics 4	User behavior analytics	Page views, events, conversions
Hotjar	Heatmaps & session recordings	Landing pages, checkout flow

13. File Structure

```

grammaretamol-student-interface/
├─ app/                                # Next.js App Router
│  └─ (auth)/                          # Auth route group (no nav/footer)
│     └─ login/page.tsx
│     └─ register/page.tsx
│     └─ forgot-password/page.tsx
│     └─ verify-email/page.tsx
│  └─ (public)/                        # Public pages (with nav/footer)
│     └─ page.tsx                      # Landing page
│     └─ courses/page.tsx
│     └─ courses/[slug]/page.tsx
│     └─ services/page.tsx
│     └─ services/[slug]/page.tsx
│     └─ about/page.tsx
│     └─ blog/page.tsx
│     └─ blog/[slug]/page.tsx

```



```

|   |   └─ faq/page.tsx
|   |   └─ contact/page.tsx
|   |   └─ testimonials/page.tsx
|   └─ (student)/          # Protected student routes
|   |   └─ dashboard/page.tsx
|   |   └─ my-courses/page.tsx
|   |   └─ my-courses/[courseId]/page.tsx
|   |   └─ live-classes/page.tsx
|   |   └─ live-classes/[id]/join/page.tsx
|   |   └─ coaching/page.tsx
|   |   └─ coaching/[bookingId]/page.tsx
|   |   └─ notifications/page.tsx
|   |   └─ profile/page.tsx
|   |   └─ transactions/page.tsx
|   |   └─ settings/page.tsx
|   └─ checkout/
|   |   └─ [courseId]/page.tsx
|   |   └─ success/page.tsx
|   └─ api/                # Next.js API routes (proxies, webhooks)
|   └─ layout.tsx          # Root layout (providers, fonts, metadata)
|   └─ globals.css         # Tailwind directives + custom properties
|   └─ loading.tsx        # Global loading UI
|
└─ components/
  |   └─ layout/           # Navbar, Footer, Sidebar, PageHeader
  |   └─ ui/              # Reusable primitives (Button, Input, Modal, Toast)
  |   └─ course/          # CourseCard, CourseGrid, CurriculumAccordion
  |   └─ learning/        # VideoPlayer, LessonSidebar, ProgressTracker
  |   └─ dashboard/       # StatCard, WelcomeBanner, ContinueLearning
  |   └─ checkout/        # OrderSummary, PaymentForm, CheckoutSuccess
  |   └─ forms/           # Form wrappers, validation schemas
  |
└─ hooks/                 # Custom React hooks
  |   └─ useAuth.ts
  |   └─ useCourses.ts
  |   └─ useProgress.ts
  |   └─ useNotifications.ts
  |   └─ useMediaQuery.ts
  |   └─ useVideoPlayer.ts
  |   └─ useProgressSync.ts
  |
└─ stores/                # Zustand stores
  |   └─ authStore.ts
  |   └─ uiStore.ts
  |   └─ cartStore.ts
  |
└─ lib/                   # Utilities, constants, types
  |   └─ api.ts           # Axios instance + interceptors
  |   └─ utils.ts         # cn() helper, formatters
  |   └─ constants.ts     # App-wide constants
  |   └─ types/           # TypeScript interfaces

```

```
|   |   | auth.ts
|   |   | course.ts
|   |   | user.ts
|   |   | payment.ts
|
| public/                                # Static assets
| | images/
| | icons/
| | fonts/
|
| styles/
| | animations.css                      # Custom keyframes
|
| middleware.ts                        # Auth guard, route protection
| next.config.js
| tailwind.config.ts
| tsconfig.json
| package.json
```

14. Security Considerations

- **XSS Prevention:** All user-generated content sanitized via DOMPurify; React's built-in escaping for JSX
- **CSRF Protection:** SameSite cookies, CSRF tokens for state-changing operations
- **Clickjacking:** X-Frame-Options: DENY, CSP frame-ancestors directive
- **Content Security Policy:** Strict CSP headers allowing only trusted script sources
- **Sensitive Data:** No API keys or secrets exposed in client bundles; env vars prefixed with NEXT_PUBLIC_ only for non-sensitive values
- **Payment Security:** PCI DSS compliance via tokenized payment flows (never handle raw card data)

End of Document